A CANONICAL ISOMORPHISM BETWEEN THE DETERMINANT OF THE OUTER TERMS AND THE DETERMINANT OF THE MIDDLE TERM OF A SHORT EXACT SEQUENCE OF FINITE LENGTH MODULES, WITH THREE COMPATIBILITY PROPERTIES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02PA (sources/stacks-chow/pinned-original-source.tex line 16203).

Allowed standard prerequisites (editorial).

- Admissible sequences, their relations, and the generation of det_kappa by a single symbol. Same chapter, chow.tex, Section on periodic complexes and determinants, in particular the relation (c) used in the sign computation at proof lines 16354-16362. Supplies the three admissible relations of det_kappa and the presentation of det_kappa by symbols; the transfer of relations and the ordered application of relation (c) are carried out in the counted span.
- Generation of det_kappa of a finite length module by one appropriate symbol. Same chapter, Lemma labelled lemma-determinant-dimension-one (source line 15850) and Lemma labelled lemma-dimension-at-most-one (source line 15761), both invoked at proof lines 16306 and 16322. Supplies that det_kappa(L) is generated by any single symbol attached to an admissible sequence generating L, which is what makes gamma surjective; the construction of gamma and the verification of its three properties stay inside the counted span.

Source reference. Original lemma 02P7: sources/stacks-chow/pinned-original-source.tex, source lines 15761–15761. Same-chapter pointer into the author environment lemma at source line 15761; invoked at proof line 16307 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02P9: sources/stacks-chow/pinned-original-source.tex, source lines 15850–15850. Same-chapter pointer into the author environment lemma

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at source line 15850; invoked at proof line 16322 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $(R, \mathfrak{m}, \kappa)$ be a local ring. For every short exact sequence*

$$0 \to K \to L \to M \to 0$$

of finite length R-modules there exists a canonical isomorphism

$$\gamma_{K \to L \to M} : \det_\kappa(K) \otimes_\kappa \det_\kappa(M) \longrightarrow \det_\kappa(L)$$

defined by the rule on nonzero symbols

$$[e_1, \ldots, e_k] \otimes [\overline{f}_1, \ldots, \overline{f}_m] \longrightarrow [e_1, \ldots, e_k, f_1, \ldots, f_m]$$

with the following properties:

(1) *For every isomorphism of short exact sequences, i.e., for every commutative diagram*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K & \longrightarrow & L & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle v} & & \downarrow{\scriptstyle w} & & \\ 0 & \longrightarrow & K' & \longrightarrow & L' & \longrightarrow & M' & \longrightarrow & 0 \end{array}$$

with short exact rows and isomorphisms u, v, w we have

$$\gamma_{K' \to L' \to M'} \circ (\det_\kappa(u) \otimes \det_\kappa(w)) = \det_\kappa(v) \circ \gamma_{K \to L \to M},$$

(2) *for every commutative square of finite length R-modules with exact rows and columns*

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & D & \longrightarrow & E & \longrightarrow & F & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & G & \longrightarrow & H & \longrightarrow & I & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

the following diagram is commutative

$$\begin{array}{ccc} \det_\kappa(A) \otimes \det_\kappa(C) \otimes \det_\kappa(G) \otimes \det_\kappa(I) & \xrightarrow[\gamma_{A \to B \to C} \otimes \gamma_{G \to H \to I}]{} & \det_\kappa(B) \otimes \det_\kappa(H) \\ \downarrow{\scriptstyle \epsilon} & & \downarrow{\scriptstyle \gamma_{B \to E \to H}} \\ & & \det_\kappa(E) \\ & & \uparrow{\scriptstyle \gamma_{D \to E \to F}} \\ \det_\kappa(A) \otimes \det_\kappa(G) \otimes \det_\kappa(C) \otimes \det_\kappa(I) & \xrightarrow{\gamma_{A \to D \to G} \otimes \gamma_{C \to F \to I}} & \det_\kappa(D) \otimes \det_\kappa(F) \end{array}$$

where ϵ is the switch of the factors in the tensor product times $(-1)^{cg}$ with $c = length_R(C)$ and $g = length_R(G)$, and

(3) *the map $\gamma_{K \to L \to M}$ agrees with the usual isomorphism if $0 \to K \to L \to M \to 0$ is actually a short exact sequence of κ-vector spaces.*

Proof. The significance of taking nonzero symbols in the explicit description of the map $\gamma_{K\to L\to M}$ is simply that if $(e_1,\ldots,e_l)$ is an admissible sequence in K, and $(\overline{f}_1,\ldots,\overline{f}_m)$ is an admissible sequence in M, then it is not guaranteed that $(e_1,\ldots,e_l,f_1,\ldots,f_m)$ is an admissible sequence in L (where of course $f_i\in L$ signifies a lift of $\overline{f}_i$). However, if the symbol $[e_1,\ldots,e_l]$ is nonzero in $\det_\kappa(K)$, then necessarily $K=\langle e_1,\ldots,e_k\rangle$ (see proof of Lemma 02P7), and in this case it is true that $(e_1,\ldots,e_k,f_1,\ldots,f_m)$ is an admissible sequence. Moreover, by the admissible relations of type (b) for $\det_\kappa(L)$ we see that the value of $[e_1,\ldots,e_k,f_1,\ldots,f_m]$ in $\det_\kappa(L)$ is independent of the choice of the lifts f_i in this case also. Given this remark, it is clear that an admissible relation for $e_1,\ldots,e_k$ in K translates into an admissible relation among $e_1,\ldots,e_k,f_1,\ldots,f_m$ in L, and similarly for an admissible relation among the $\overline{f}_1,\ldots,\overline{f}_m$. Thus γ defines a linear map of vector spaces as claimed in the lemma.

By Lemma 02P9 we know $\det_\kappa(L)$ is generated by any single symbol $[x_1,\ldots,x_{k+m}]$ such that $(x_1,\ldots,x_{k+m})$ is an admissible sequence with $L=\langle x_1,\ldots,x_{k+m}\rangle$. Hence it is clear that the map $\gamma_{K\to L\to M}$ is surjective and hence an isomorphism.

Property (1) holds because

$$\begin{aligned} & \det_\kappa(v)([e_1,\ldots,e_k,f_1,\ldots,f_m]) \\ = \;& [v(e_1),\ldots,v(e_k),v(f_1),\ldots,v(f_m)] \\ = \;& \gamma_{K'\to L'\to M'}([u(e_1),\ldots,u(e_k)]\otimes[w(f_1),\ldots,w(f_m)]). \end{aligned}$$

Property (2) means that given a symbol $[\alpha_1,\ldots,\alpha_a]$ generating $\det_\kappa(A)$, a symbol $[\gamma_1,\ldots,\gamma_c]$ generating $\det_\kappa(C)$, a symbol $[\zeta_1,\ldots,\zeta_g]$ generating $\det_\kappa(G)$, and a symbol $[\iota_1,\ldots,\iota_i]$ generating $\det_\kappa(I)$ we have

$$\begin{aligned} & [\alpha_1,\ldots,\alpha_a,\tilde{\gamma}_1,\ldots,\tilde{\gamma}_c,\tilde{\zeta}_1,\ldots,\tilde{\zeta}_g,\tilde{\iota}_1,\ldots,\tilde{\iota}_i] \\ = \;& (-1)^{cg}[\alpha_1,\ldots,\alpha_a,\tilde{\zeta}_1,\ldots,\tilde{\zeta}_g,\tilde{\gamma}_1,\ldots,\tilde{\gamma}_c,\tilde{\iota}_1,\ldots,\tilde{\iota}_i] \end{aligned}$$

(for suitable lifts $\tilde{x}$ in E) in $\det_\kappa(E)$. This holds because we may use the admissible relations of type (c) cg times in the following order: move the $\tilde{\zeta}_1$ past the elements $\tilde{\gamma}_c,\ldots,\tilde{\gamma}_1$ (allowed since $\mathfrak{m}\tilde{\zeta}_1\subset A$), then move $\tilde{\zeta}_2$ past the elements $\tilde{\gamma}_c,\ldots,\tilde{\gamma}_1$ (allowed since $\mathfrak{m}\tilde{\zeta}_2\subset A+R\tilde{\zeta}_1$), and so on.

Part (3) of the lemma is obvious. This finishes the proof. □